

CHAPTER 2

PERCEPTION (VISUAL)



OVERVIEW

1. General introduction
2. Eye Movements (and reading)
3. Interaction between bottom-up and top-down processes

SENSING THE WORLD

- Our environment changes continuously and rapidly
 - some changes are potentially harmful or beneficial
 - our species has evolved perceptual mechanisms to identify such relevant information.
- We do not have direct access to what is happening in the environment
 - we build an internal representation of the external world
 - representations of reality are not the same as reality.

INTRODUCTION CHAPTER 2



SENSATION AND PERCEPTION

- **Sensors** (a.k.a. **receptors**)
 - specific sensory neurons that capture a physical property
 - e.g. photoreceptors are neurons sensitive to light.
- **Transduction**
 - translation of physical stimulation into signals that are interpretable by the brain.
- **Sensations** are the building blocks of perception
- **Perceptions** are elaborations of sensations
 - sensations and perceptions cannot exist on their own
 - Together, they form our sensory experience of the world.

SENSATION VS PERCEPTION

- Perception = more than just "conscious" perception
 - it goes through many steps beyond our awareness
 - interpretation usually very fast -unconscious-
- Consciousness seems added in later stage
- This is evident in the case of problems with visual perception
- Some examples
 - Face blindness: <http://www.youtube.com/watch?v=VKa-PuJCrO4>
 - Hemispatial neglect:
<https://www.youtube.com/watch?v=d4FhZs-m7hA>
& <http://www.youtube.com/watch?v=ADchGO-0kGo>
 - Visual agnosia:
<https://www.youtube.com/watch?v=ze8VVtBgK7A>

HUMAN PERCEPTUAL SYSTEMS

Vision	The ability to detect electromagnetic radiations within a narrow band of frequencies (visible light)
Audition	The perception of air vibrations
Taste	The detection of various chemical compounds in food and liquids
Smell	Similar to taste, but detects chemical compounds in the air
Touch	The perception of pressure on skin
Equilibrioception	The perception of balance
Thermoception	The perception of temperature
Nociception	The perception of pain

SENSATION VS PERCEPTION



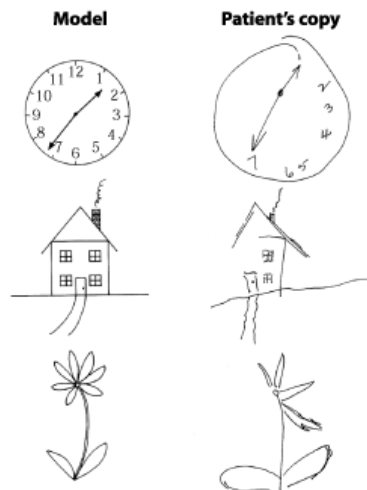
VISUAL AGNOSIA AND PROSOPAGNOSIA

- Visual agnosia
 - lesions occur in the inferior region of temporal cortex that selectively impact the ability to recognize objects
 - able to extract a reasonably intact perception of what they see but are unable to assign any meaning to this percept.
- Prosopagnosia
 - the recognition of faces is severely impaired after brain damage
 - but can recognize the identity of individuals by properties such as voice, hairstyle or glasses.

HEMINEGLECT

- Hemineglect:
 - Most frequently associated with a lesion of the right parietal lobe
 - Result is a deficit in attention to and awareness of one side of space
 - Eating only half of the plate
 - Not writing “underpants” but “pants”
 - = they do not “see” that something is missing

HEMINEGLECT



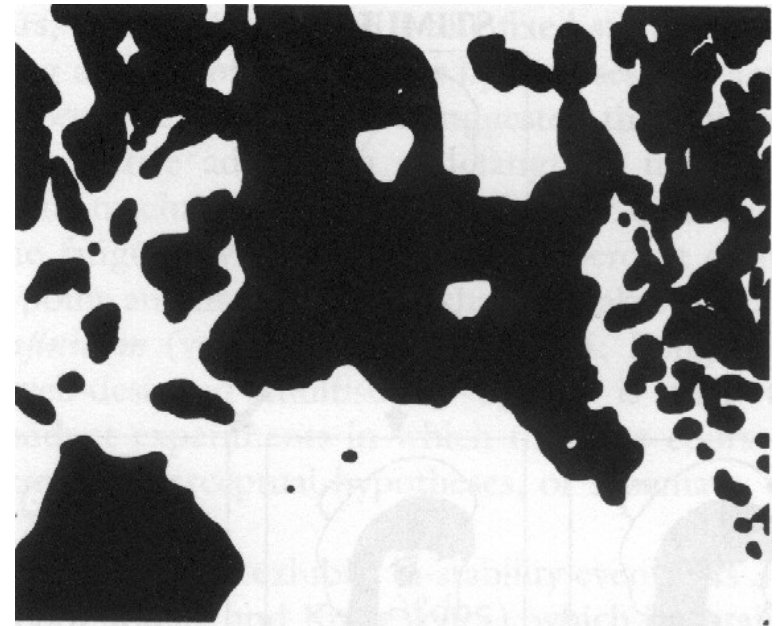
SENSATION VS PERCEPTION



SENSATION VS PERCEPTION

- Sensations are with both pictures like they always are:
 - Blots of black and white
- However, there is a difference in interpretation/organisation/...:
 - Perception takes/needs time for pictures like the the first one
 - Normally it's going quickly (see the second one)

SENSATION VS PERCEPTION



SENSATION VS PERCEPTION



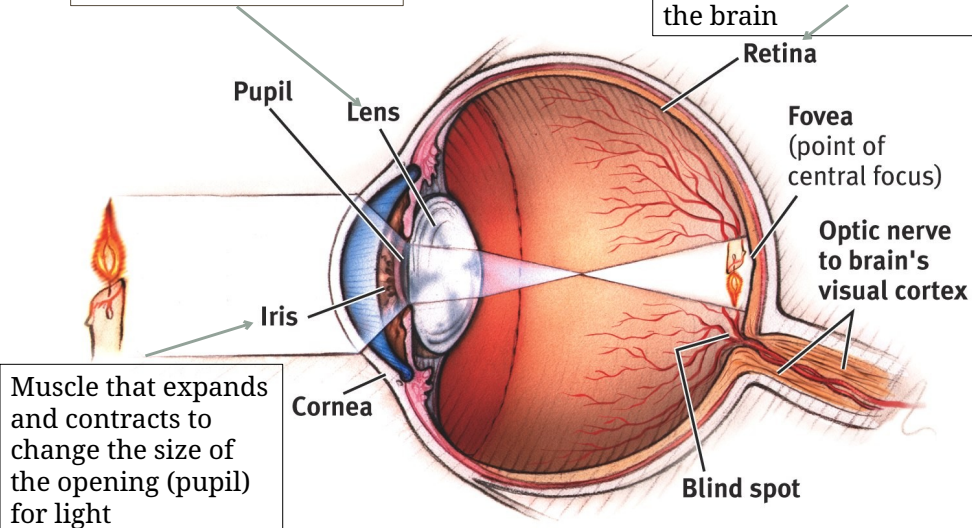
SENSATION VS PERCEPTION



THE EYE

Contains sensory receptors that process visual information and sends it to the brain

Focuses the light rays on the retina

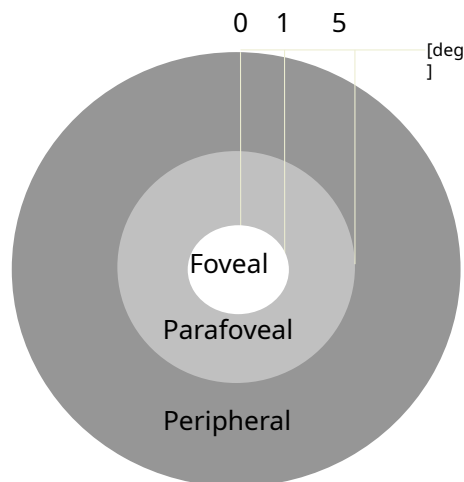


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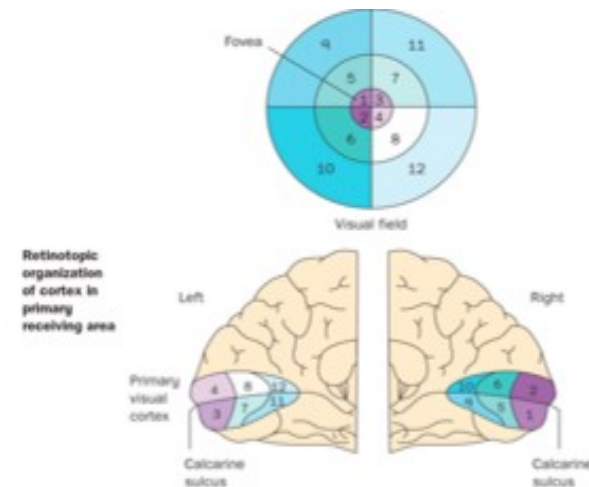
VISUAL PERCEPTION: THE EYE

- Retina:
 - Composed of fovea, parafovea and perifovea
 - Rods (94%; sensitive to light) and cones (6%; capture colors)
 - Cones are concentrated in centre of retina - the fovea
- Visual acuity:
 - greatest in fovea; drops off sharply in parafovea & perifovea.
- Peripheral vision:
 - 15-50% of acuity of fovea and also less colour-sensitive
- Human visual system works by moving eye so that info of interest falls on fovea.
 - Eyes move all the time (even during sleep)
- Visual system integrates information extracted on successive saccades.

VISUAL PERCEPTION: THE EYE



RETINOTOPIC ORGANIZATION OF VISUAL CORTEX



VISUAL INFORMATION PROCESSING

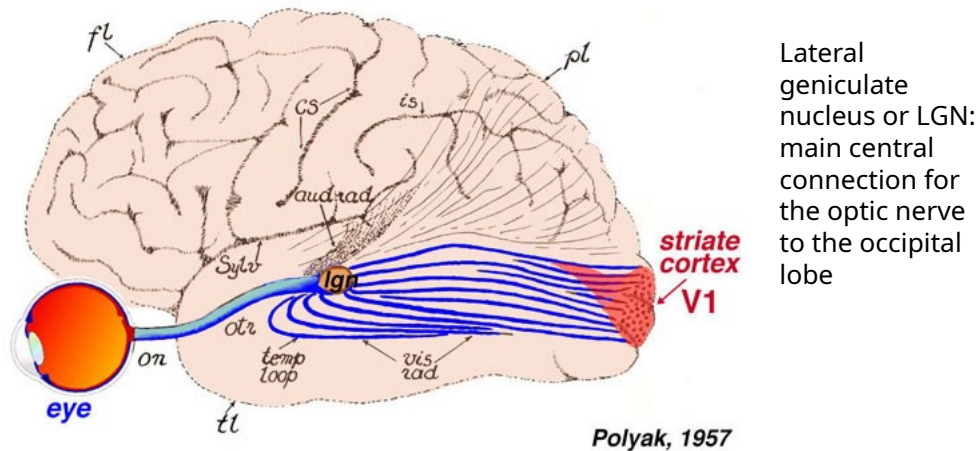
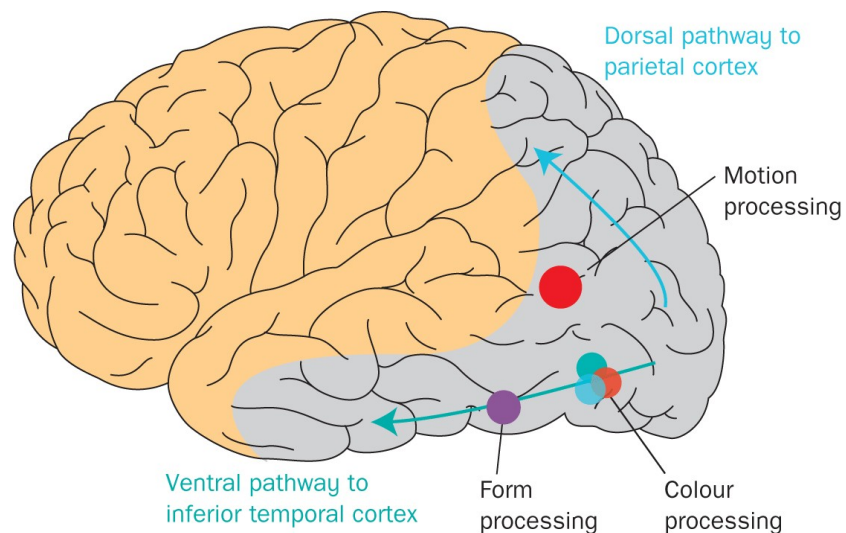


Figure 8. Visual input to the brain goes from eye to LGN and then to primary visual cortex, or area V1, which is located in the posterior of the occipital lobe. Adapted from Polyak (1957).

VISUAL SYSTEM

- Ventral stream
 - From occipital cortex to temporal cortex
 - The **'what'** pathway
 - Involved in recognition of the objects being viewed
- Dorsal stream
 - From occipital cortex to parietal cortex
 - The **'where'** pathway
 - Involved in locating and guiding how to use an object
- Same scene, two streams: E.g., garden
 - Perceiving shapes and colours of flowers and recognizing them:
 - Ventral stream
 - Swat away a wasp on your arm:
 - Dorsal stream

VISUAL SYSTEM



2. EYE MOVEMENTS (& READING)



BASIC CHARACTERISTICS OF EYE MOVEMENTS

- *Fixations:*
 - stationary target is stabilised on the fovea.
- *Pursuit eye movements:*
 - eye movement matches velocity of target to stabilise retinal image.
- *Saccadic eye movements:*
 - Eye movement to locate target onto fovea.
 - Saccadic eye movements used during reading and visual search
 - Saccades last 20-100 msec & fixations last 200-600 milliseconds

BASIC CHARACTERISTICS OF EYE MOVEMENTS

- Have been studied for many years
- Can be used to infer moment-to-moment cognitive processing in the brain
- Can be easily tracked with the current Eye Tracking technology

FIXATIONS

- Eye is (*relatively*) still and “fixated” to the certain point.
 - *E.g.* reading a single word.
- All the information from the scene is (mainly) acquired during fixation.
- Duration varies from 120-1000 ms, typically 200-600 ms.
 - Interspersed with saccades...

FIXATIONS

- Three types of eye movement occur during fixation:
 - *Tremor:*
 - high frequency movements that are not matched between eyes
 - (probably): represent *noise* in the activity of the visual system.
 - *Drift:*
 - low movement of eye from and around target
 - this is thought to represent *noise* in the visual system.
 - *Microsaccades:*
 - occur once or twice a second
 - may be small corrective eye movements to re-locate the target onto the retina.

SACCADIC EYE MOVEMENTS

- Saccades are:
 - highly accurate, high velocity eye movements
 - used to foveate objects of interest.
- Vision is composed of series of saccades providing snapshots of visual information that have to be re-integrated by the visual system.
 - During reading we make a series of saccadic eye movements to sample words.
 - During visual search task we make a series of darting saccadic eye movements to view potential targets.
- Movement is ballistic:
 - the end point of saccade cannot be changed during the movement

SACCADIC SUPPRESSION



Take a bright light and stare at it, then moved the light very quickly: the light would smear across your vision.

This effect essentially is the same as the effect obtained when using time-lapse techniques to photograph traffic.

Why doesn't the visual image smear across your vision during a saccade?

Studies suggest that visual input is suppressed during a saccade.

EASY DETECTION OF CHANGE



- In this image, you can see a change quite clearly.
 - The human visual system is very good at detecting such events.
- Even if you weren't initially fixating the region of the change with your narrow field of high-resolution vision (the fovea),
 - the low-resolution peripheral field is well suited to detecting the flicker
 - and subsequently a saccade would be executed to bring the changing area to the centre of attention.



- Next, you will see two alternating frames of an image as above,
 - but in between each image there will be a brief (100 millisecond) interruption in which the entire image turns grey.
 - Try to see if you can detect the change in the image...



- This is almost the same as the previous animation
- Difference: the intervening grey flashes have been removed.



- Knowing where the changes are lets you now detect them easily
- It's hard to believe that they couldn't be seen before.

WHAT RELEVANCE DOES CHANGE BLINDNESS HAVE?



- Is it just another example of the quirky sorts of visual illusions which psychologists love to collect?
 - After all, it's not often our vision is interrupted by 100 millisecond duration grey rectangles covering our visual field.
- In actuality, however, our vision *is* interrupted in exactly such a way
 - but our perceptual system does a lot of work to hide this from us.

WHAT RELEVANCE DOES CHANGE BLINDNESS HAVE?

- The most common behaviour performed by a human being is the saccade, or rapid eye movement.
 - During each of those movements, our vision is degraded markedly, due to the eyes rotating at hundreds of degrees per second- too rapidly for the retina to integrate useful information.
 - We execute 100 000 to 200 000 saccades every waking day, each lasting roughly 20 - 100 milliseconds.
- Hence, *we spend 10-20% of our waking lives functionally blind.*
 - That this is something of which we are not generally aware is perhaps the greatest illusion of all time, and illustrates the brain's immense capacity for self-delusion.

PURSUIT EYE MOVEMENTS

- Move your finger in front of you and follow it with your eyes
- The finger remains stationary on the retina, yet you see it moving!
- The background now seems to move relative to the stationary finger
- Vision is NOT suppressed during the movement

PURSUIT EYE MOVEMENTS

- Pursuit eye movements are used to track moving objects, e.g. following a moving car, following a tennis ball as it comes toward you, ...
- People are capable of maintaining fixation on fast moving objects
 - tennis balls, cricket balls, footballs, ...
 - when the slightest error would impair performance of a secondary task (e.g. hitting or catching the ball).
- Factors affecting efficiency of pursuit eye movements:
 - alcohol, fatigue, age, ...

SUMMARY OF EYE STRUCTURE AND FUNCTION

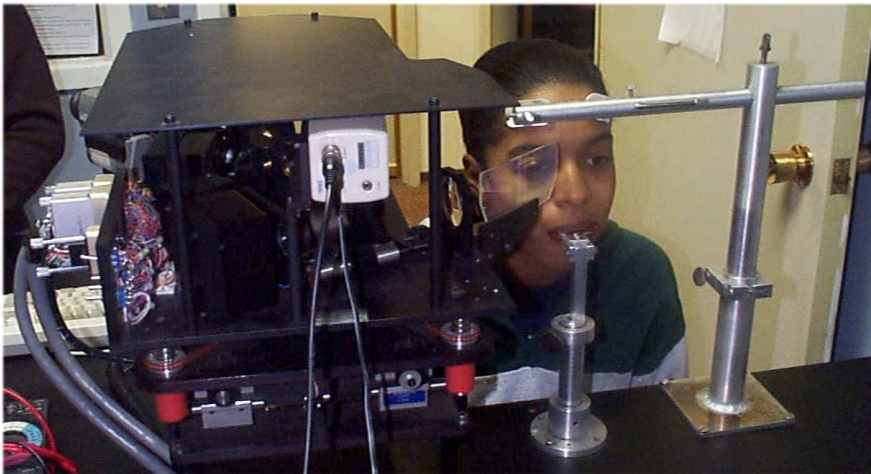
- The visual system works by shifting the eyes so that a target is superimposed on the fovea.
- There are 3 main types of eye movement:
 - During *fixation* the eye is relatively stable while information is extracted.
 - Saccade eye movements are rapid jerky movements of the eye that are used to locate the target on the fovea.
 - A combination of fixations and saccades produce a sequence of snapshots of the visual environment that must be re-integrated by the brain.
 - Pursuit eye movements enable the viewer to locate a moving object on the fovea.

MEASURING EYE MOVEMENTS

- Dual Purkinje (DPI) eye tracker
- Head mounted eye trackers
- Remote eye movement tracking equipment

DUAL PURKINJE EYE TRACKER

- Floods eye with infrared light and measures reflections from particular parts of the eye.
- Locks onto reflections from the front of the pupil (1st Purkinje reflection) and reflection from back of the lens (4th Purkinje reflection).
- Tracks position, sequence and duration of fixations made during cognitive task.
- Widely used to study reading behaviour/scene perception.



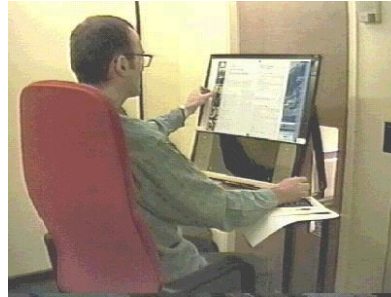
HEAD MOUNTED EYE TRACKERS

- Records video image of eye(s) and video image of direction of vision.
- System determines point of gaze of eye(s), which can be superimposed on video image of direction of vision.
- More flexible,
 - system used to assess driving behaviour
 - Used to examine interaction between spoken word recognition and visual search



REMOTE EYE MOVEMENT TRACKING EQUIPMENT

- Discrete video camera records eye pupil.
- System determines point of gaze of eye, which can be superimposed on video image of direction of vision.



2. EYE MOVEMENTS (& READING)



EYE MOVEMENT CONTROL DURING READING

- The perception of a complex scene/text involves
 - a complicated pattern of fixations, where the eye is held (fairly) still
 - and saccades, where the eye moves to foveate a new part of the scene
- The main issue is “Where to look next?” Answer **may** be that perception is influenced by
 - *Cognition*, e.g. “meaningful” and parts previewed with parafoveal vision
 - *Visual cues*; highly visually attractive areas attracted by peripheral vision

EYE MOVEMENT CONTROL DURING READING

- Skilled readers are unaware of difficulties in controlling the movement of their eyes across the page.
- However this is a complex task:
 - Ensure that eye moves from word to word without missing information.
 - Ensure that enough information is taken in on each fixation.
 - Ensure that eye lands on appropriate point in a word for it to be processed correctly.
 - Maintain fixation on word for correct period of time, and make saccade to view next word at correct time.

COUNT THE NUMBER OF 'F' IN THE FOLLOWING TEXT

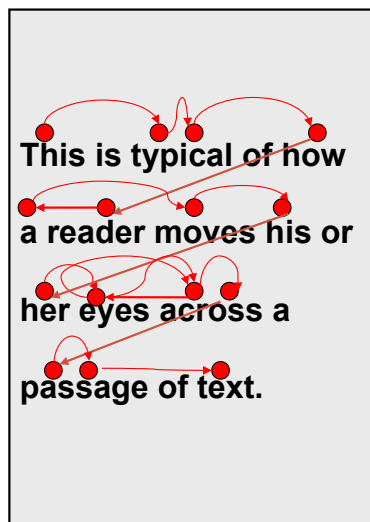
FINISHED FILES ARE THE RESULT OF YEARS OF SCIENTIFIC STUDY COMBINED WITH THE EXPERIENCE OF YEARS

EYE MOVEMENT CONTROL DURING READING

- Readers do not move their eyes smoothly across the page.
- They make a sequence of fixations separated by saccades:
- Fixations last 200-250 msec but with intra- & inter-reader variability.
- Saccade last 20-30 msec.

EYE MOVEMENT CONTROL DURING READING

- Readers do not move their eyes smoothly across the page.
- They make a sequence of fixations separated by saccades:
- Fixations last 200-250 msec but with intra- & inter-reader variability.
- Saccade last 20-30 msec.
- A reader typically makes 1 or 2 fixations on content words in text, but often skips function words (e.g. a, the, in, of, at, etc)
 - 80% of content and 20% of function words are fixated.
- About 10% of saccades are regressive saccades to re-inspect earlier words in text.



EYE MOVEMENT CONTROL DURING READING

- What amount of information is taken in during a fixation (this is called the perceptual span)?
- Rayner & colleagues used 'moving window' paradigm to determine size of perceptual span.
- Eye movements are monitored as subject reads text, but only part of currently fixated text is visible.
- What size of text window is needed for normal reading?

Xxxx xxxx xxx xxxx xxx xxxx
xxxx switched xxxxxxxx xxxxxx.
Xxxx xxxxxxx xxxxxx xx xxxxxx.



Xxxx xxxx xxx xxxx xxx xxxx
xxxx xxxxxxx positions xxxxxx.
Xxxx xxxxxxx xxxxxx xx xxxxxx.

EYE MOVEMENT CONTROL DURING READING

- Perceptual span: subjects read normally when window includes text 3-4 letters to left of fixation, and 14-15 letters to right of fixation.



- Perceptual span is asymmetric.
 - Asymmetric to the right for English, French, Dutch, ... readers.
 - Asymmetric to the left for Hebrew readers
 - Smaller for Japanese readers (13 letters)
- Chinese: smaller than Japanese span (12 letters)
 - Influence of reading skill:
 - Beginning readers smaller span (12 letters to the right)
 - Still asymmetric to the right

SOME OTHER READING DATA

Approximate mean fixation time	250 milliseconds
Range of fixation time	100-500 millisecond
Average progressive saccades	7-9 characters ~ 2 visual degrees
Average regressive saccades	3-4 characters ~ less than 1 visual degree
Average Percentage of regressive saccades	10-20% of all saccades

For dyslexic and poor readers:

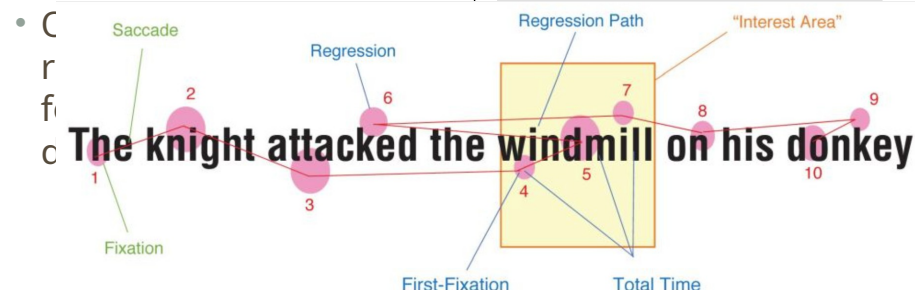
- Longer fixation durations
- Shorter and larger number of saccades
- More regressive saccades

Changes in Reading with Age

	Grade												
	1	2	3	4	5	6	7	8	9	10	11	12	C
Average duration of fixations (in seconds)	0.33	0.30	0.28	0.27	0.27	0.27	0.27	0.27	0.27	0.26	0.26	0.25	0.24

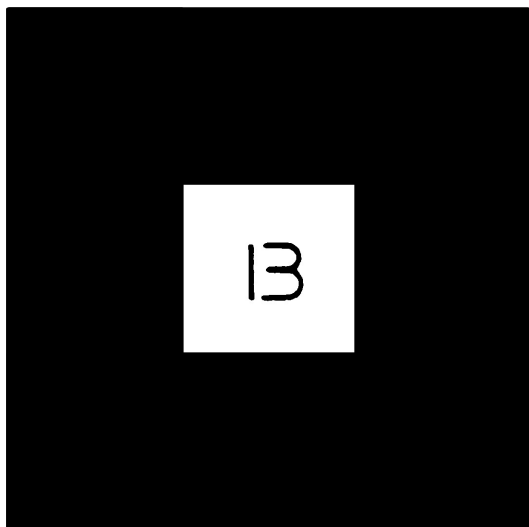
EYE MOVEMENT CONTROL DURING READING

- Often make regressive fixations within word to correct for incorrect initial landing position.



EYE MOVEMENT CONTROL DURING READING: THEORIES

- Occulomotor models:
 - Decisions of when and where to move the eyes are primarily determined by *low-level* factors
 - Hence, decisions to move the eyes are determined
 - By visual properties of text (word lenght, spaces,)
 - Limitations in visual acuity
 - E.g. Strategy-tactics model of O'Regan



EYE MOVEMENT CONTROL DURING READING: THEORIES

- Problems for the occulomotor models :
 - Readers are more likely to refixate on low-frequency words than on high-frequency words
 - (laundry, apology, trim, ...) vs (the, have, man, year, because, thing, ...)
 - Words with lower frequency and unfamiliar words need longer fixations
 - Words with higher frequency and familiar words need shorter fixations
 - Fixation time of a word is longer when it is preceded by a rare word (spillover effect)
 - Detecting particular letters is less accurate in highly familiar words
 - Proofreading is harder the more familiar the text

EYE MOVEMENT CONTROL DURING READING: THEORIES

- Processing models:
 - Fixations are influenced by lexical factors and moment-to-moment comprehension processes
 - Occulomotor factors influence eye movements, but less than cognitive factors
 - E.g., E-Z reader (Reichle, Pollatsek, Fisher, & Rayner, 1998)

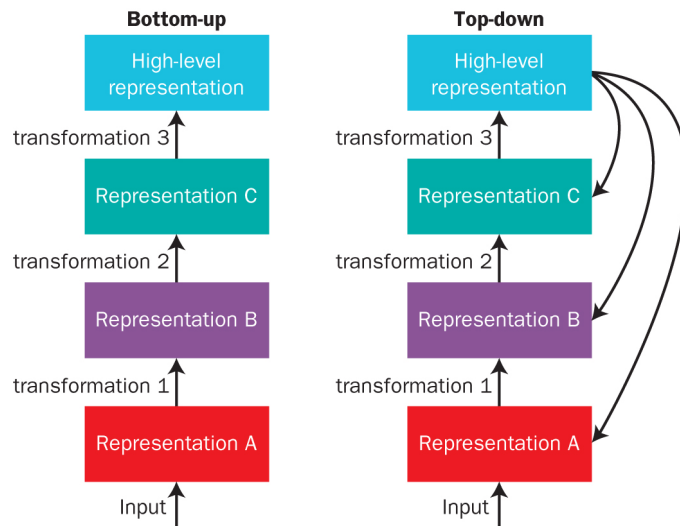
3. BOTTOM-UP AND TOP-DOWN PROCESSES



BOTTOM-UP AND TOP-DOWN PROCESSES IN PERCEPTION

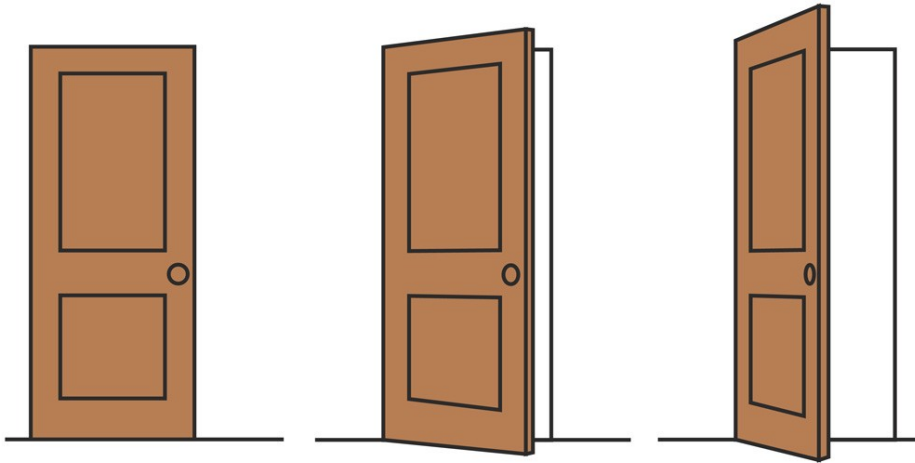
- Information can be/is ambiguous
 - the brain uses previous knowledge to remove ambiguity.
- *Bottom-up processing or stimulus-driven processing*
 - from sensory information up to conceptual information.
- *Top-down processing or concept-driven processing*
 - high-level cognition controls and regulates events affecting lower levels of cognition
 - may use context, expectations and knowledge to structure information
 - influence of top-down processes is limited; some bottom-up assumptions cannot be corrected.

INFORMATION FLOW



BOTTOM-UP AND TOP-DOWN PROCESSES IN PERCEPTION

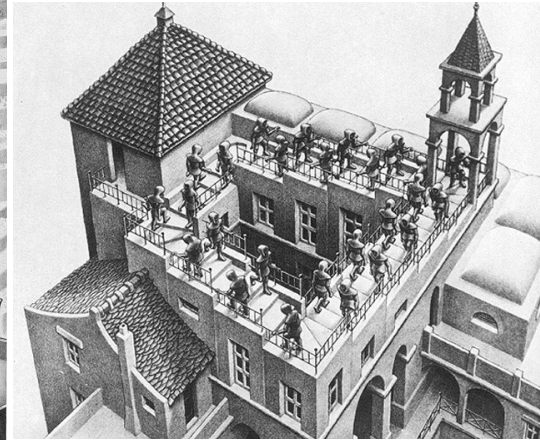
- Visual input forces the visual system to work with partial environmental information.
 - Occlusion
 - Limitations of viewing angle
 - Limited sampling frequency of sensors
- Brain must go beyond the sensory data in providing us with an accurate representation of environment
 - Brain *constantly* constructs hypotheses based on the input
 - Tests these hypotheses
 - Reject weak, adopt the strongest $\Rightarrow$ stable state



STABILITY VS PLASTICITY

- Brain must have:
 - Stability: To maintain continuity across time
 - Plasticity: To deal with unexpected inputs
- Most theories:
 - Formation of perceptual hypotheses
 - Continuous interaction between schema-driven and data-driven mechanisms

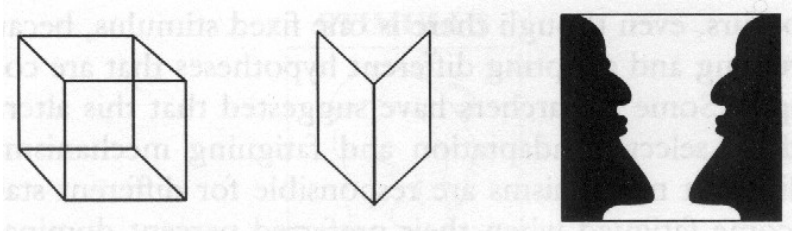
MAURITS CORNELIS ESCHER



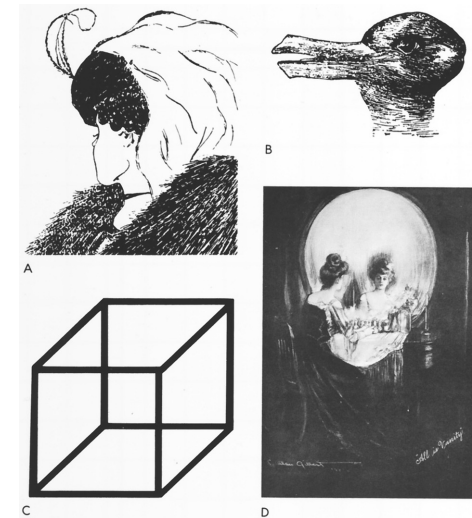
BOTTOM-UP AND TOP-DOWN PROCESSES IN PERCEPTION

- Often much redundant information:
 - Reality appears stable (=dominant perceptual state), because only weak alternative hypotheses
- In special occasions, this constant search and testing of hypotheses can be observed.
- Evidence from multistable figures:
 - Hypothesis-constructing-and-testing process does not stop after arriving at a stable state

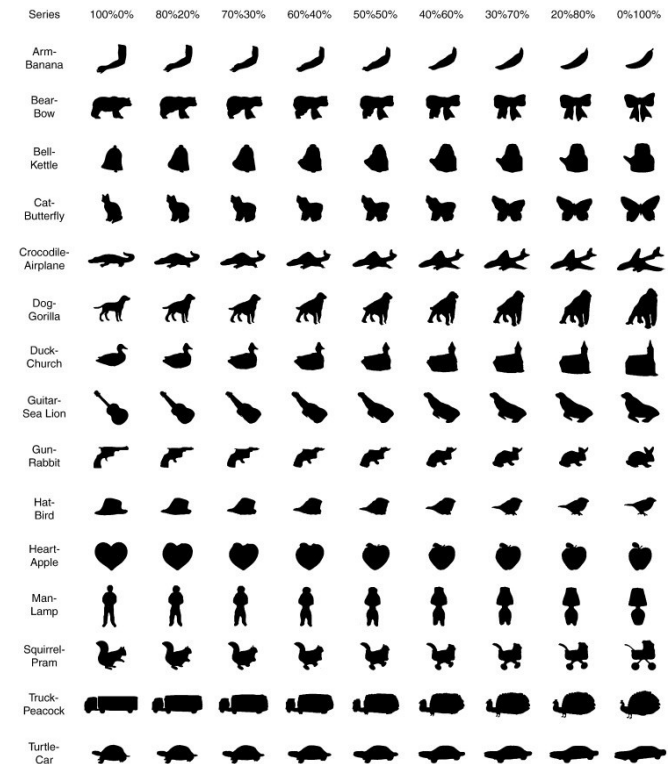
EVIDENCE FOR INTERACTION: MULTISTABLE FIGURES



OTHER MULTISTABLE FIGURES



DIFFERENT VERSIONS OF MULTISTABLE FIGURE



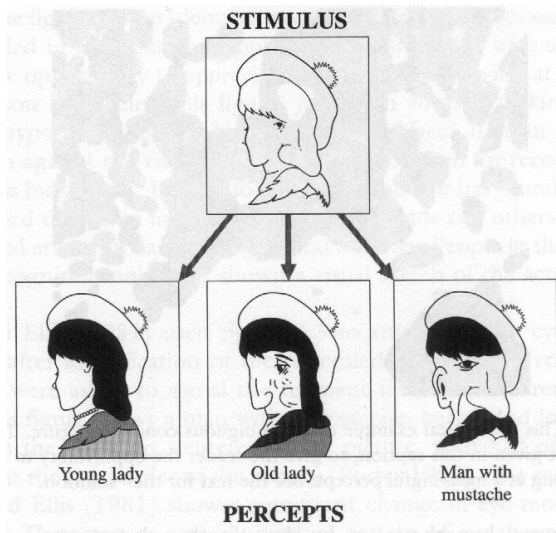
STILL ANOTHER AMBIGUOUS FIGURE



OBSERVERS' BEHAVIOR

- If you can look long at the multistable figures:
 - Spontaneous reversals among stable percepts
 - One interpretation is adopted, the other suppressed
 - Then the other adopted
 - Continuous alternations
 - Alternation occurs, even though there is one fixed stimulus:
 - Because brain keeps constructing and adopting different hypotheses that are consistent with visual input
 - Possibility: fatiguing mechanisms

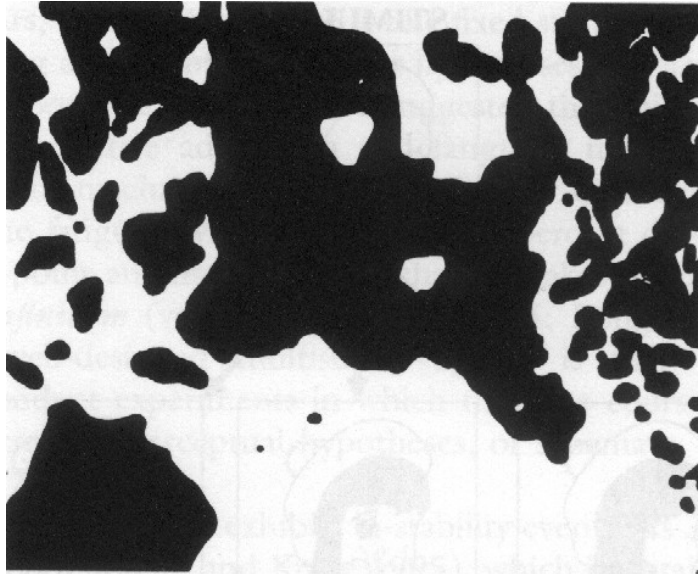
TRI-STABLE AMBIGUOUS FIGURE AND ROLE OF FIXATIONS



RESULTS

- Fixational pattern during the time of any given percept was correlated to the corresponding perceptual hypothesis
- Saccadic eye movements determined by interaction between perceptual hypothesis and stimulus characteristics

OTHER EVIDENCE: AMBIGUOUS CONCEALED FIGURES



LIU BOLIN



OBSERVER'S BEHAVIOR

- Significant change in eye movements after recognition
 - Nearly random pattern of eye movements during 75 sec preceding identification
 - A well-recognized scanpath that visits the salient features of the face during the 75 sec following recognition
- Evidence that cognitive, schema-driven models direct eye movements

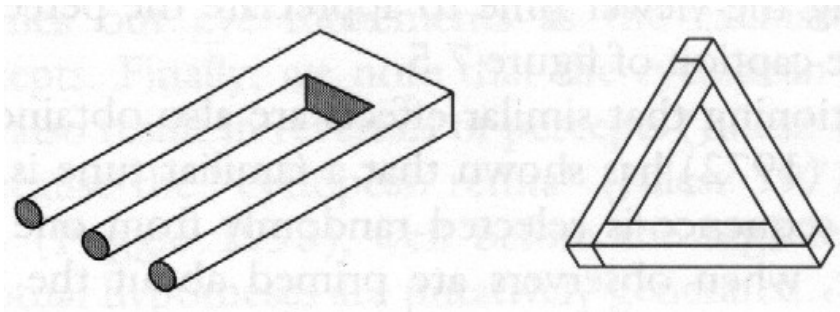
LIU BOLIN



LIU BOLIN

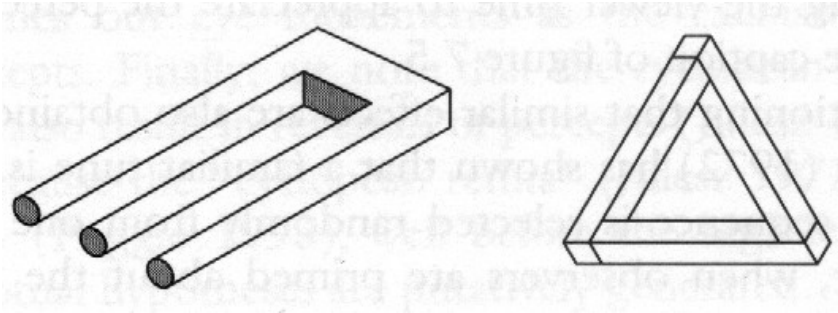


OTHER EVIDENCE: IMPOSSIBLE FIGURES

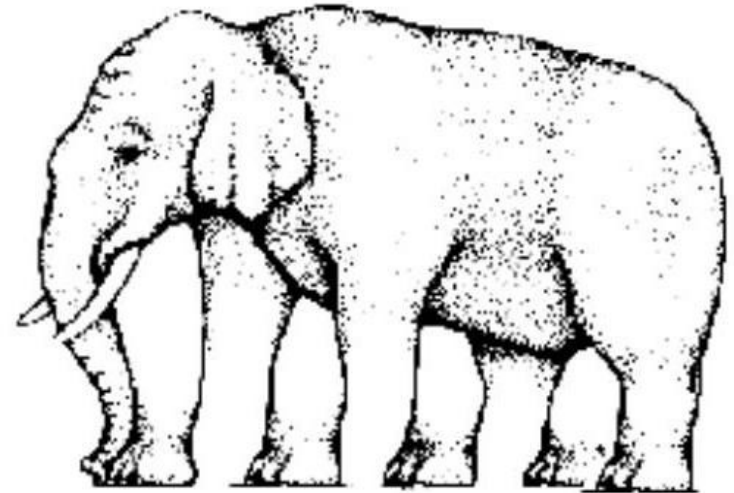


- Different from previous examples:
 - Entire image accounts for percept
 - With parts perceived as background
- Perception of impossible figures
 - Depend on the focal point

OTHER EVIDENCE: IMPOSSIBLE FIGURES



EIGHT-LEGGED ELEPHANT



BALCONY OR PATIO?



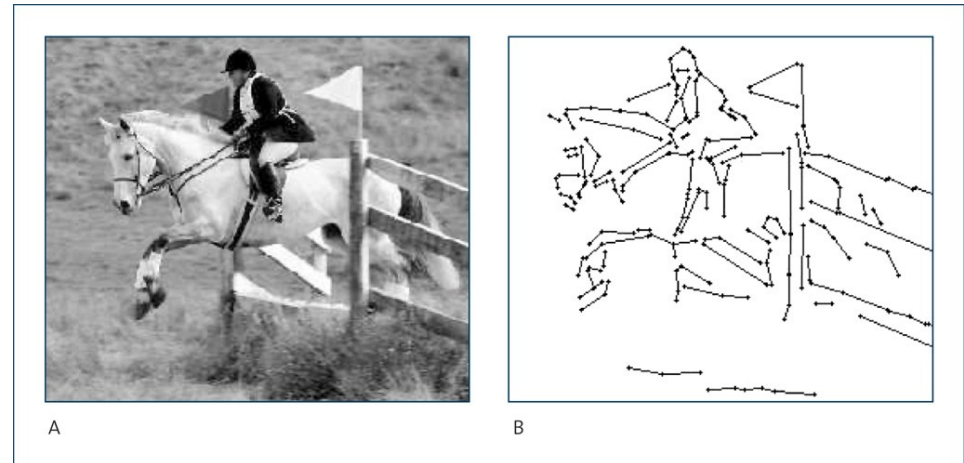
BOTTOM-UP AND TOP-DOWN PROCESSES IN PERCEPTION

- Interaction is necessary to explain previous findings
 - One could argue:
 - Unambiguous S: bottom-up processes dominate
 - Ambiguous S: top-down processes dominate
 - But, for many situations the two can be seen to work together
 - Sugase et al., 1999:
 - Individual visual neuron activity in monkeys
 - *Global* info (human faces vs. monkey faces vs. shapes) was encoded and transmitted faster than *fine* info about identity or expression.
 - This global info, which is transmitted ahead in the processing stream, seems to work as a "header", to prepare destination areas for more detailed information.
- = example of top-down influence on mid-level processing of facial information

MARR'S (1982) THEORY

- How are complex objects recognised so quickly?
- Marr developed a computer model of object recognition.
- Theory assumes *three bottom-up stages*
 1. First Level
 - starting from a grey-scale representation, contours of objects are identified
 - contours are used to identify edges and surfaces
 - following the Gestalt rules, edges are grouped together to form tokens.

MARR'S (1982) THEORY: PRIMAL SKETCH



GESTALT PRINCIPLES

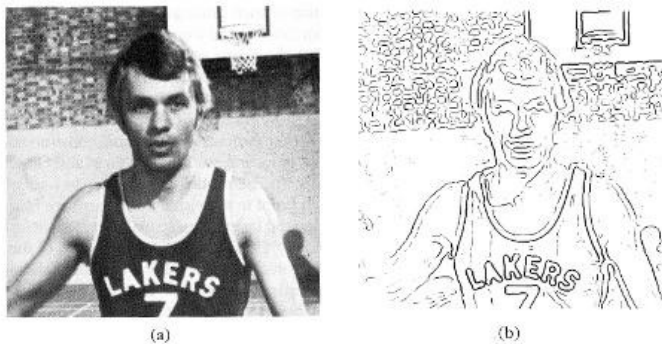
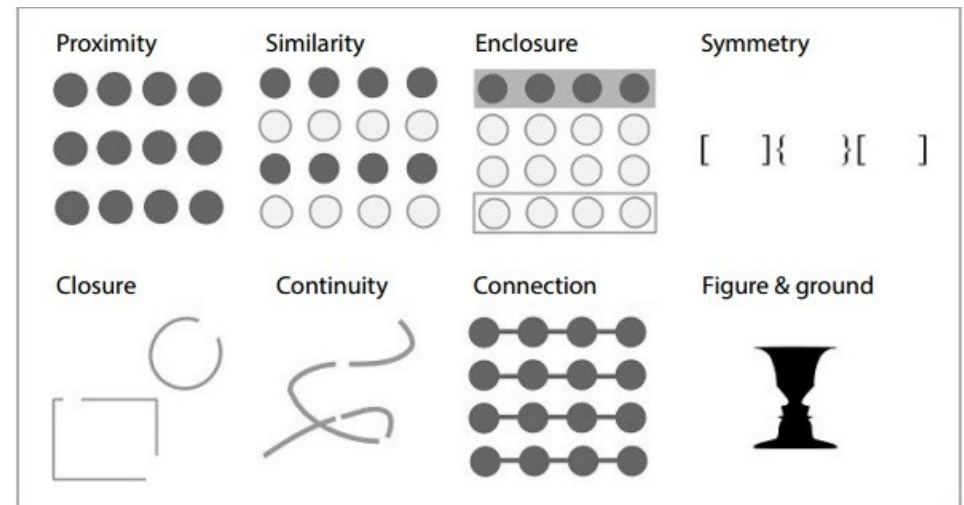
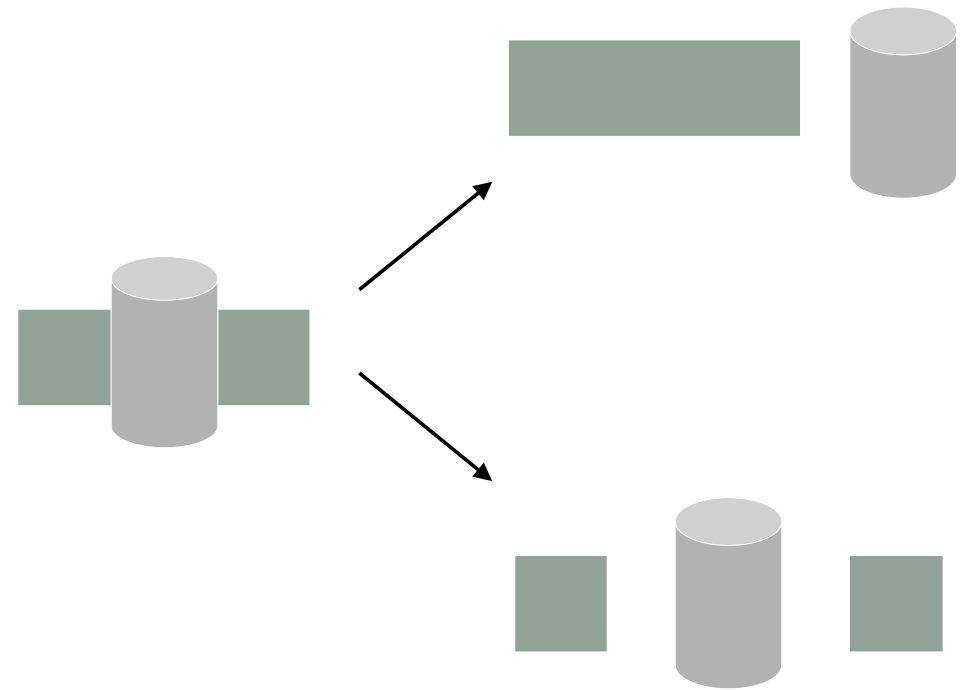


Figure 2-15. Another example of zero-crossings, here, the intensity of the lines has been made to vary with the slope of the zero-crossing, so that it is easier to see which lines correspond to the greater contrast. (Courtesy BBC Horizon.)



GESTALT PRINCIPLES

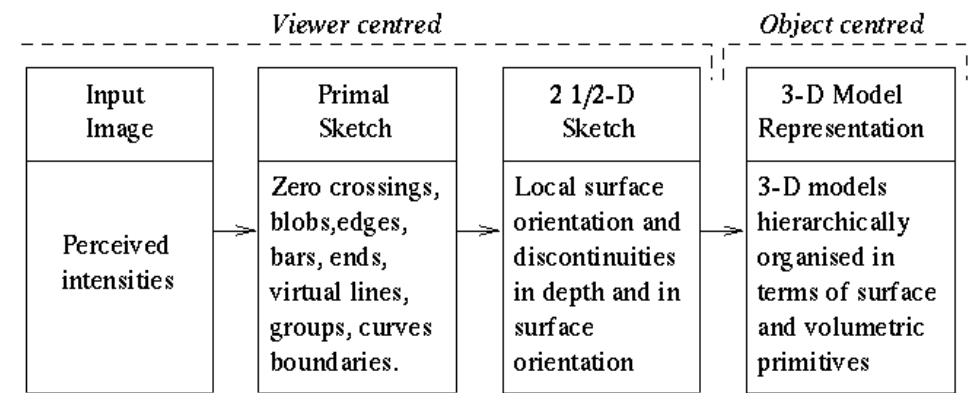
- **Proximity:**
 - We see three rows of dots instead of four columns of dots because they are closer horizontally than vertically.
- **Similarity:**
 - We see similar looking objects as part of the same group.
- **Enclosure:**
 - We group the first four and last four dots as two rows instead of eight dots.
- **Symmetry:**
 - We see three pairs of symmetrical brackets rather than six individual brackets.
- **Closure:**
 - We automatically close the square and circle instead of seeing three disconnected paths.
- **Continuity:**
 - We see one continuous path instead of three arbitrary ones.
- **Connection:**
 - We group the connected dots as belonging to the same group.
- **Figure & ground:**
 - We either notice the two faces, or the vase. Whichever we notice becomes the figure, and the other the ground



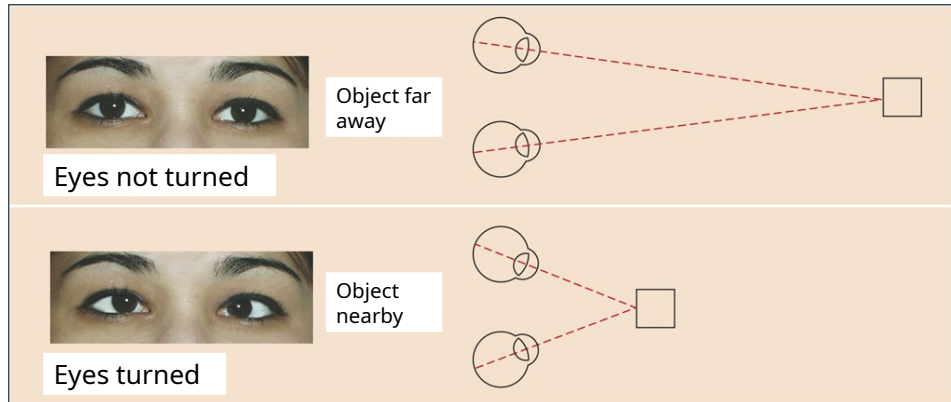
MARR'S THEORY (II)

2. Second level – choice of representation
 - the third dimension is incorporated
 - the *range map* adds information about distance to every point of the two-dimensional (2D) representation
 - distance and angle are measured from the point of view of the perceiver.

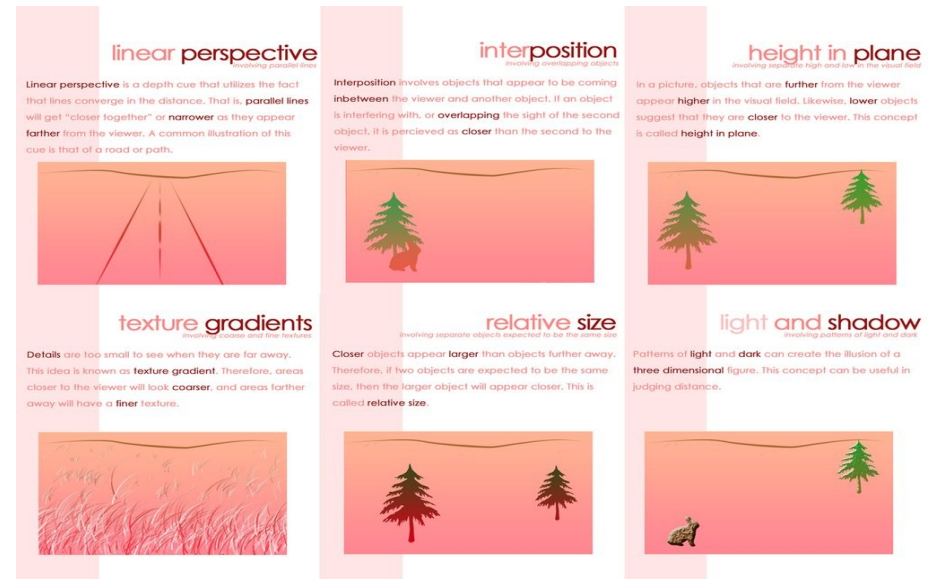
MARR IN A FIGURE



BINOCULAR DEPTH CUES



MONOCULAR DEPTH CUES



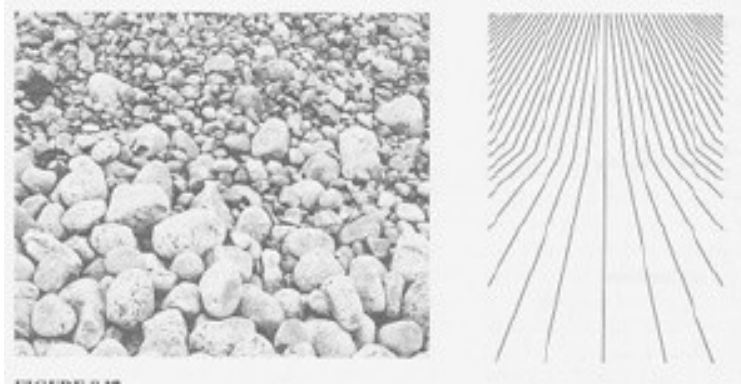
RELATIVE SIZE



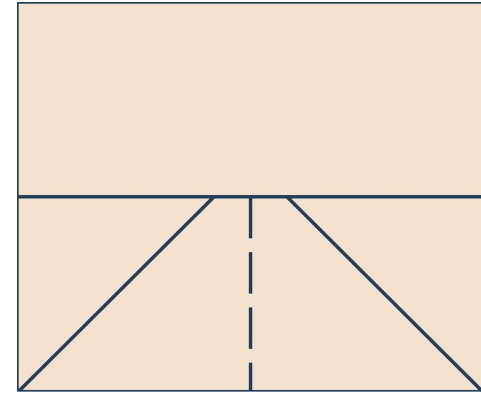
RELATIVE SIZE



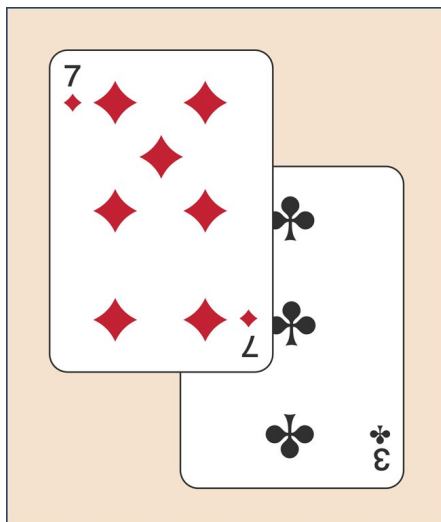
TEXTURE GRADIENTS



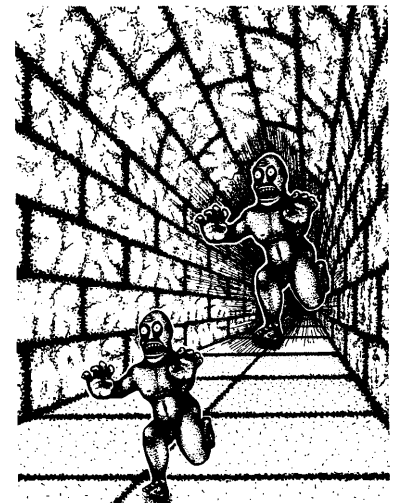
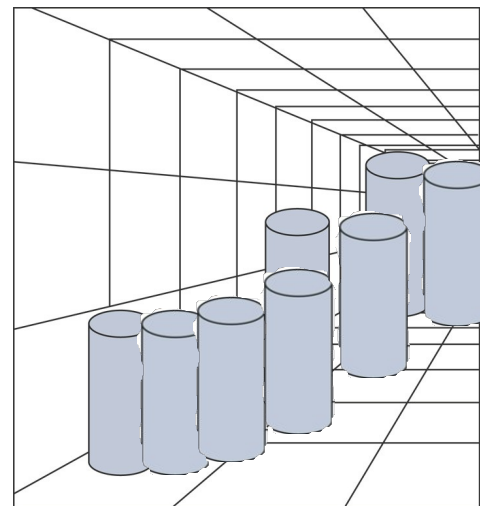
LINEAR PERSPECTIVE



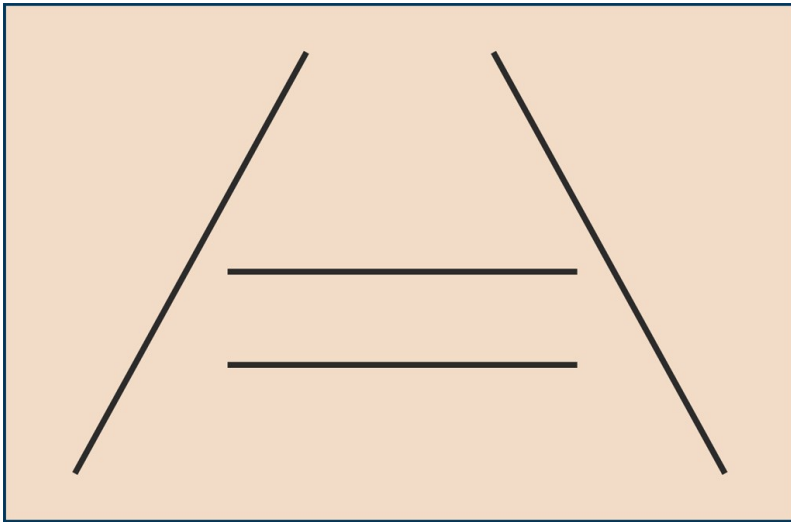
INTERPOSITION



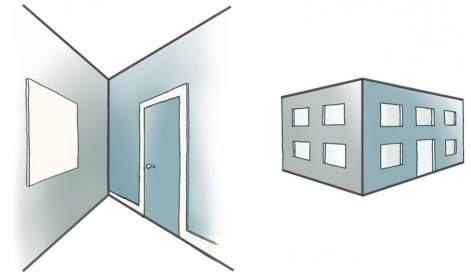
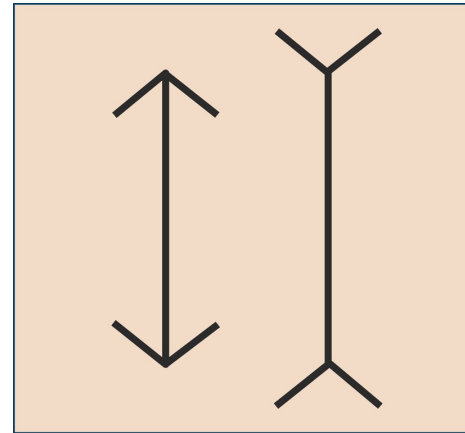
SOME ILLUSIONS



SOME ILLUSIONS



SOME ILLUSIONS



SOME ILLUSIONS



MARR'S THEORY (II)

2. Second level – choice of representation
 - the third dimension is incorporated
 - the range map adds information about distance to every point of the two-dimensional (2D) representation
 - distance and angle are measured from the point of view of the perceiver.
3. Third level – achieving the computations
 - matching of the target object with an internal representation
 - there is a move from a view-dependent representation to a view-independent representation.

RECOGNITION

- Different shapes of an object but one identity:



OBJECT RECOGNITION

- Template matching:
 - One template
 - However, too many different shapes of the same object
 - (see the letter A)
- Feature analysis
 - Deconstructing an object into a set of component features than can be compared with a library
 - See Selfridge, 1958

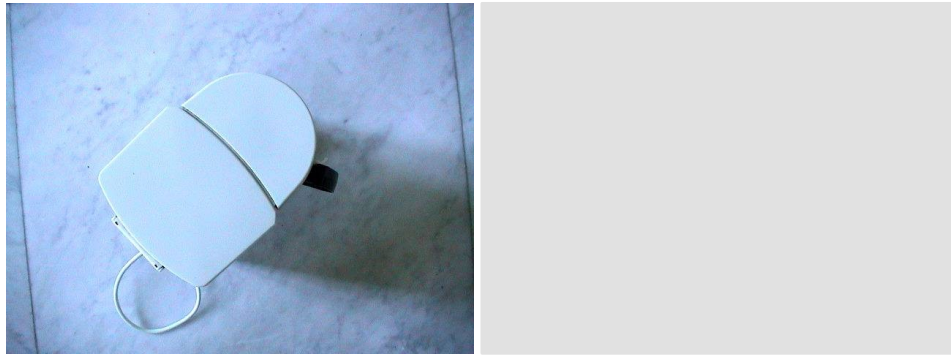
OBJECT RECOGNITION

- Prototype theory
 - Using the most typical example of a category, see e.g. Rosch (1973)
 - Prototype is easy to recognize, the more distance, the harder
 - E.g. bird = beak + feathers + fly
 - Robin is more typical than penguin
 - Hierarchy within categories:
 - Basset hound = dog (basic category) = mammal = animal

OBJECT RECOGNITION

- Visual object recognition
 - viewpoint invariant relationships are important:
 - Any aspects of an object that is preserved no matter the direction from which we view the object

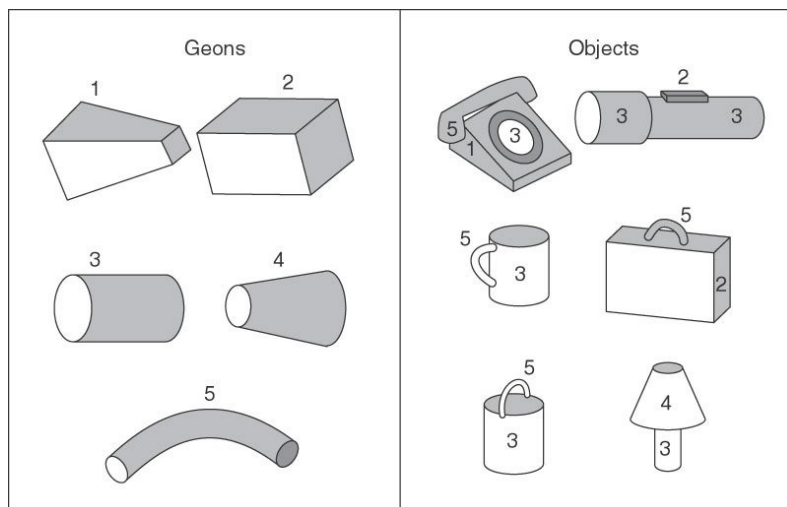
MOVE FROM VIEW-DEPENDENT TO VIEW-INDEPENDENT REPRESENTATION(2,5D VS 3D)



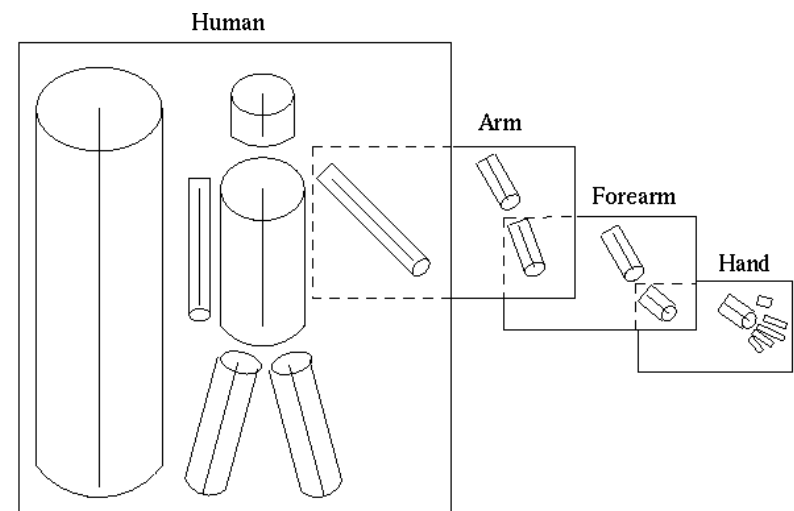
OBJECT RECOGNITION

- Visual object recognition
 - viewpoint invariant relationships are important:
 - Any aspects of an object that is preserved no matter the direction from which we view the object
 - Biederman and his 36 geons (inspired by Marr):
 - *Volumetric primitives* than can be recognized from any viewpoint
 - RBC: Recognition by Components theory
 - Experiment:
 - Line drawings of objects
 - Parts deleted
 - Performance suffered substantially when the regions containing viewpoint invariant information were removed

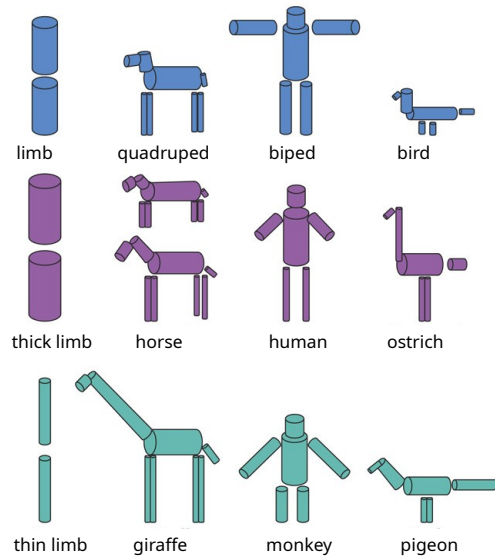
BIEDERMAN AND 36 GEONS



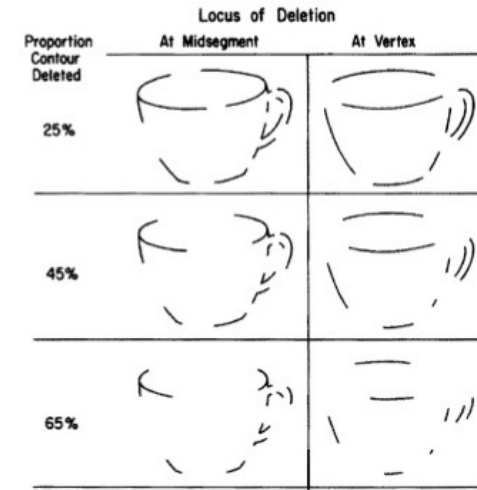
THE 3-D MODEL REPRESENTATION IN MARR'S FRAMEWORK



THE 3-D MODEL REPRESENTATION IN MARR'S FRAMEWORK



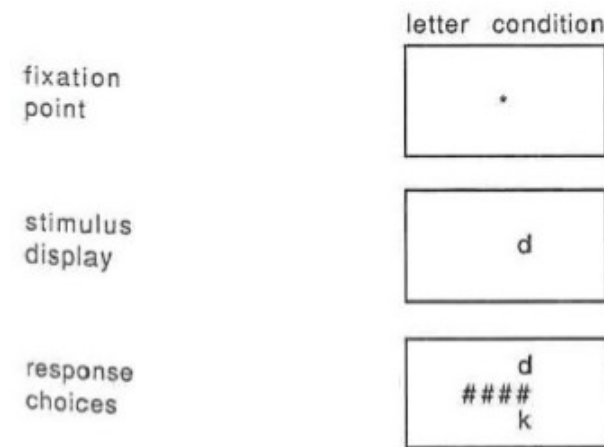
OBJECT RECOGNITION



OBJECT RECOGNITION

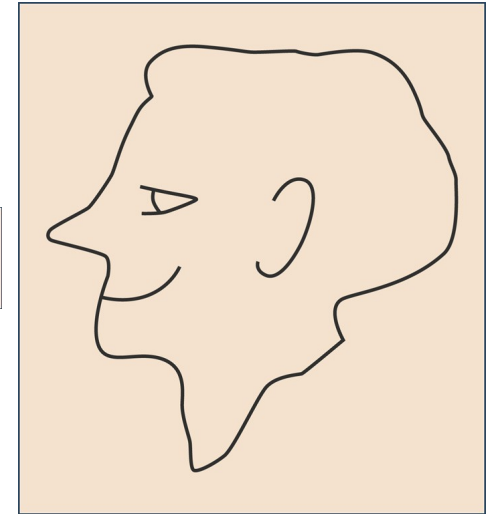
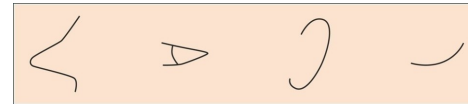
- Nowadays:
 - We seem to use both view-independent properties and view-dependent properties
 - Hence, more experience with views from certain angle has an influence
- Shape is crucial for Human Object Recognition:
 - E.g. colour "only" comes into play when shape is ambiguous
 - LTM-colour-knowledge can play a role in top-down mechanisms

WORD- AND OBJECT-SUPERIORITY



WORD- AND OBJECT-SUPERIORITY

Aoccdrnig to a rscheearch at Cmabrigde Uinervtisy, it deosn't mttar in waht oredr the ltteers in a wrod are, the olny iprmoetnt tihng is taht the frist and lsat ltteer be at the rghit pclae. The rset can be a toatl mses and you can sitll raed it wouthit porbelm. Tihs is bcuseae the huamn mnid deos not raed ervey lteter by istlef, but the wrod as a wlohe.



WORD- AND OBJECT-SUPERIORITY



WORD- AND OBJECT-SUPERIORITY

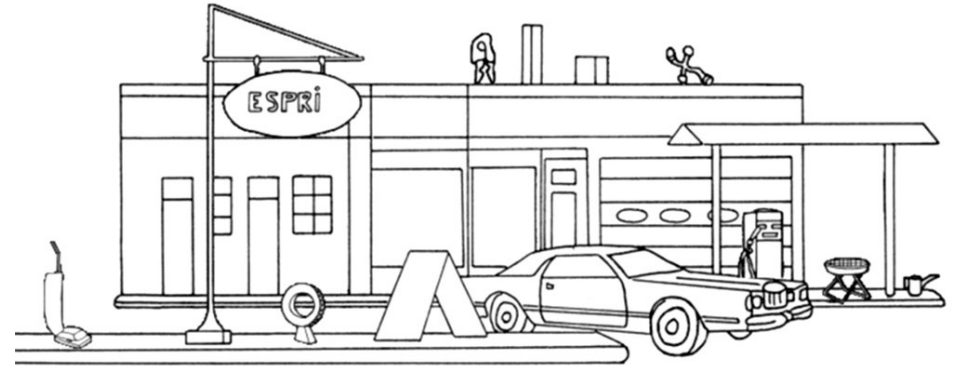


SCENES

- Perception of an environment
 - Natural extension of studies in object recognition
- Humans are very good at rapidly processing visual scenes.
 - E.g. example of visual scanning of a scene



SCENE-SUPERIORITY



- “Count the number of non-objects”:
 - in order to force the participants to scan the complete scene
- Unexpected objects are fixated longer

PERCEPTION OF BIOLOGICAL MOTION

- We are very good in using human action to identify identity, gender, emotion, the action carried out, ...
- Even with very little information available, we are good in recognising biological motion:
 - Point-light walkers of Johansson, 1973
 - <http://www.biomotionlab.ca/Demos/BMLwalker.html>
 - <http://www.patakosmos.com/bio-motion-lab/>

PERCEPTION AND ACTION

- In the past: first perception, then action
 - Consequence: perception and action are separate processes
- Nowadays: more and more demonstrations that both processes are not independent

MIRROR NEURONS

- Rizzolatti
 - Research with monkeys, single cell recordings in motoric cortex
- Video
 - <http://www.youtube.com/watch?v=XzMqPYfeA-s&feature=related>
 - <http://www.youtube.com/watch?v=xmEsGQ3JmKg&feature=related>



MIRROR NEURONS

- <http://www.youtube.com/watch?v=XzMqPYfeA-s&feature=related>



MIRROR NEURONS

- Are neurons with the special property that they represent both the sensory aspects of perceiving actions as well as motor aspects of how to produce the action
- According to some scientists, they provide the basis for understanding the goals of others:
 - a malfunctioning mirror-neuron system could be behind the social isolation of autism

BUT

- *The Myth of Mirror Neurons: The Real Neuroscience of Communication and Cognition*, by Gregory Hickok
- “Every now and again an idea from science escapes from the lab and takes on a life of its own as an explanation for all mysteries, a validation of our deepest yearnings, and irresistible bait for journalists and humanities scholars...”

BUT ...

- Hickok puts an end to this monkey business by showing that mirror neurons do not, in fact, explain language, empathy, society, and world peace.
- But this is not a negative exposé—the reader of this book will learn a great deal of the contemporary sciences of language, mind, and brain, and will find that the reality is more exciting than the mythology.